

Question: Is the Energy Required for AI Due to Its Inherent Inefficiency?

Posted by u/MarvinBEdwards01 • 0 points • 13 comments

My impression is that these AI data centers are putting pressure on the electrical infrastructure. And that this may be due to the fact that they are answering questions using innately inefficient algorithms.

Could we accomplish an energy reduction by creating specialized AIs, where the human can be steered to the most efficient machines based upon the nature of their question?

For example, we could dedicate an AI to looking up things in a set of encyclopedias, or looking for answers about television, music, and theater.

The notion that an AI is trying to predict the next word in its response based on its prior words, word by word, sounds like a very inefficient (and energy expensive) way to do its work.

Comments

u/Mandoman61 • 1 points • 2026-02-21 14:24 UTC

Inefficiency is certainly a factor.

Yes, companies are actively working on efficiency.

But this gets offset by demand. Even if you reduce cost by 50% if you service 150% more users you have a net increase in resource usage.

u/MarvinBEdwards01 • 1 points • 2026-02-21 15:50 UTC

I just had a chat with ChatGPT. The thing in the back of my mind was the Soundex system that was used to recall similar sounding names with different spellings by assigning a phonetic code. My notion was that similar questions, which already had an answer, could be looked up if the questions were encoded in a condensed form that would map to an answer previously computed.

u/MarvinBEdwards01 • 1 points • 2026-02-21 15:19 UTC

But with new users you seldom get new questions. So, in most cases the work has already been done at least once before. So, how do we find the old answer, to avoid doing the same work over and over?

u/Mandoman61 • 1 points • 2026-02-21 15:35 UTC

Yes they are probably cashing common answers. But this still requires processing time.

u/[deleted] • 1 points • 2026-02-18 14:45 UTC

[removed]

u/loosingkeys • 1 points • 2026-02-18 14:03 UTC

Yes, the current way that LLMs work is very energy-intensive. You could probably call it "inefficient", but this is the only way humans have been able to create anything close to AI.

It's being made more efficient by huge margins all the time. But the base underlying technology it is built on is still incredibly computationally intense.

u/VelvetOnion • 1 points • 2026-02-18 03:45 UTC

There is a push, by some, that thermodynamic computing will be the answer to AI energy use. But anything to that claims to be the alternative to quantum computing and the revolution for AI energy use that I don't understand I can't vouch for.

If it is real, some of the calculations would be quicker because AI algorithms would be better suited to the new hardware paradigm. Both AI and thermodynamic computer is non-deterministic. If it is real you'd still need to wait for hardware production to scale to a critical mass, the speed to scale to faster than the equivalent current approach and then there would be a bunch of rework to move the models across the new platform.

Sounds cool but so many ifs.

u/collin-h • 1 points • 2026-02-18 03:37 UTC

even if they were super efficient.... answering a million variants of the same stupid questions from all of us stupid people will inherently inject inefficiency into the system. The efficient way to handle that would be to write the answer down in one place and let everyone who asks it reference the same answer... but we already had that with google and other search engines. This is generating bespoke answers to the same dumb questions billions of times over.

u/Remote-College9498 • 1 points • 2026-02-18 03:16 UTC

It is a thermodynamic problem. Basically, with AI you create knowledge in a short time compared to the human brain that used thousands of years. Furthermore, the AI machine must be in a kind of non-equilibrium to create new knowledge (knowledge in an abstract sense), constantly needing power (i. e. over time that is energy). In thermodynamic words the AI machine must run at non-zero epistemological entropy. If it was zero, the AI machine would be in a crystallized state, that means no new knowledge is generated (and power consumption becomes theoretically zero) or saying having no intelligence anymore to create new knowledge. In such a state the AI machine is like an encyclopedia, a book-like, and the energy is just used to look up and transfer the information of the knowledge to the user, e. g. Google engine.

u/SgathTriallair • 1 points • 2026-02-18 02:57 UTC

AI isn't very costly in terms of energy. It is stressing the grid because it's a new demand on a system that isn't growing.

In terms of computer uses, AI is lower on the tier list. The only exception to this is the initial training but that is only done once per model so if you amortize it to the entire model lifecycle it's still small.

u/inteblio • 1 points • 2026-02-18 01:58 UTC

Yes! You are exactly right.

But people are lazy, the costs are relatively small compared to the usefulness.

It'll get more efficient.

But maybe think of it like this- it would use the maximum compute anyway.

u/paraffin • 1 points • 2026-02-18 05:08 UTC

Yup. Jevon's paradox. If we make more efficient AI then that increases the use cases for it and we end up using more power, not less.

u/Eyshield21 • 1 points • 2026-02-18 00:54 UTC

part of it is just scale. the other part is we're still in the "throw compute at it" phase. efficiency will matter more when margins get squeezed.

u/Comfortable-Web9455 • 2 points • 2026-02-18 00:54 UTC

I don't think you quite understand the complexity of human language, with respect. Can you imagine how much information is required to cross reference every single word in English with every other word in English and in a way that encodes how the words can be used. The amount of processing power required to learn enough to function as an AI and the amount of processing power required to answer even a simple AI prompt is astronomical. The maths involved is absurdly complex.

Of course there will be better ways of doing this. But right now we don't have the hardware, we're bumping up against the limits of what physics allows in chips, and we do not have the computational techniques to do any better with the technology we have at this time.

This is a problem the AI world is working on. But we don't have anything better right now. What we have now is literally as sophisticated as the human race can build.

u/MarvinBEwards01 • -1 points • 2026-02-18 03:43 UTC

>Can you imagine how much information is required to cross reference every single word in English with every other word in English and in a way that encodes how the words can be used.

Yeah. In SQL it is called a "Cartesian Product", and is to be avoided except with very small sets of data.

>The amount of processing power required to learn enough to function as an AI and the amount of processing power required to answer even a simple AI prompt is astronomical.

Exactly. That's the problem. It is not an efficient way to solve the problem.

>The maths involved is absurdly complex.

And if you do the math one word at a time it becomes absurdly inefficient. Ideally, the AI would have a separate math interpreter, as if the AI had a calculator in its hand. That would be much more efficient.

>What we have now is literally as sophisticated as the human race can build.

We need to do better. Either that or every data center should have its own source for generating its own electricity.

u/SadSeiko • 0 points • 2026-02-18 12:49 UTC

The human brain runs on 20W and is considerably more complex than an LLM. People in here are talking about the more complex it becomes the more energy it uses but that's not even the case with us. Most of our brain is dedicated to keeping us moving and not even thinking

u/Comfortable-Web9455 • 2 points • 2026-02-18 09:33 UTC

I am sorry but you don't understand how LLMs work if you think it is even vaguely like an SQL database or that architecturally maths can be separated into an interpreter level like a calculator. An LLM is maths. You are treating it like words are held as strings to which mathematical processes are applied. It is not like that. The raw data is vectors held in huge multi-dimensional vector matrixes. Take away the maths and there is nothing left.